COS30018 - Intelligent System

Project: Stock Price Prediction System

Project Summary

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1. Introduction

Predicting financial market movements is a notoriously complex challenge due to high volatility, non-linear patterns, and noise present in time-series data. This project presents a comprehensive, multi-phase machine learning pipeline designed to model stock price movements using sequence-based deep learning models, hybrid ensembling, and multi-modal financial news sentiment analysis.

The project is structured into two main components: Regression & Ensemble Forecasting (Tasks C.1–C.6) and an Advanced Extension via Classification & Sentiment Analysis (Task C.7).

The initial phase establishes a robust technical infrastructure and explores time-series regression models using historical Open-High-Low-Close-Volume (OHLCV) market data:

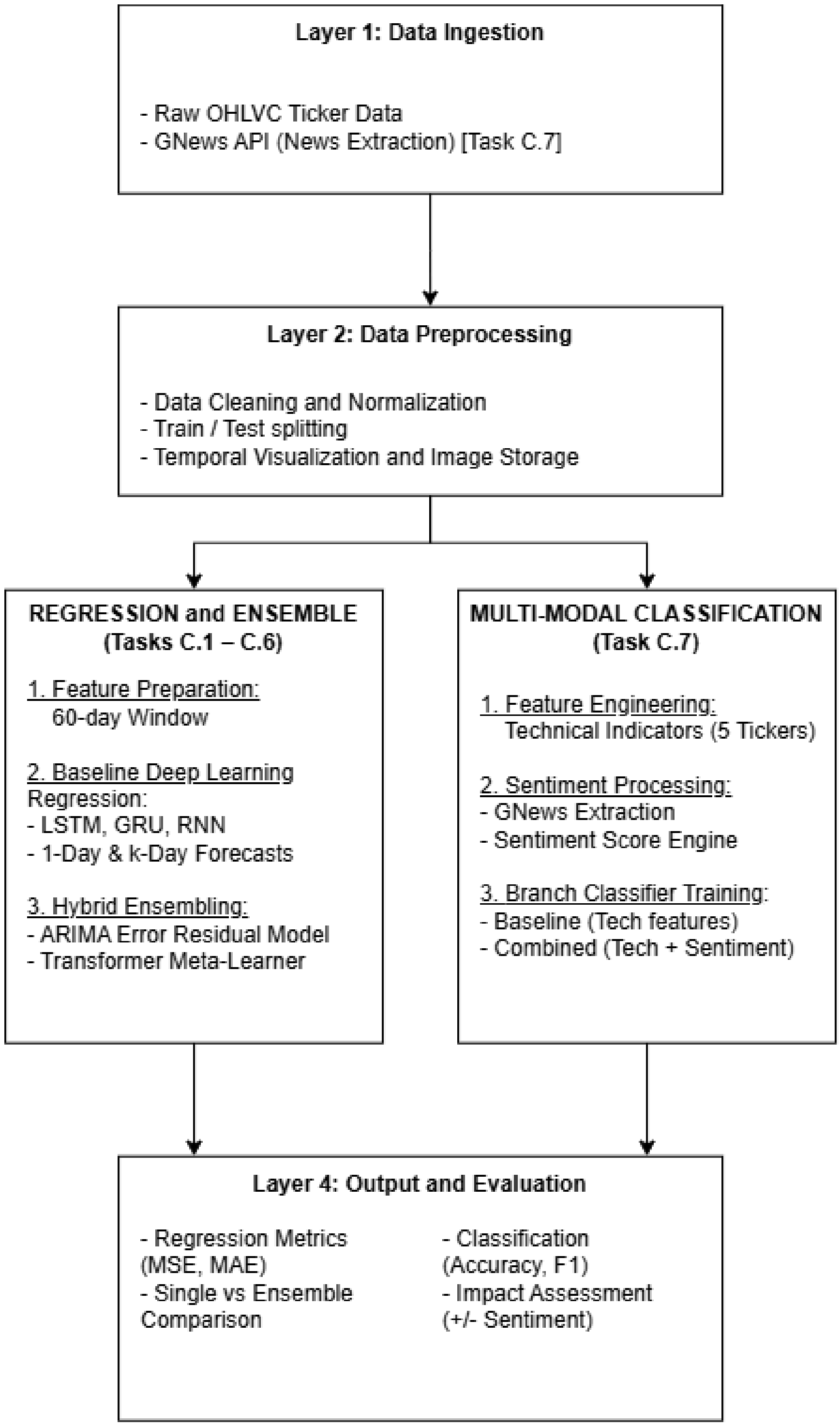
- **Environment & Pipeline Setup (Tasks C.1–C.3):** Establishes the local development environment, GitHub version control, data splitting strategies (train/test), and data visualization frameworks, including converting temporal data to images for prospective spatial processing.
- **Deep Learning Modeling (Tasks C.4–C.5):** Implements foundational recurrent neural networks—including Recurrent Neural Networks (RNN), Gated Recurrent Units (GRU), Long Short-Term Memory (LSTM), and Feedforward networks. Using a rolling window of 60 historical trading days, these architectures are configured for both single-step (1-day ahead) and multi-step (k-days ahead) multivariate stock forecasting.
- **Advanced Ensemble Approaches (Task C.6):** Enhances prediction stability and accuracy by integrating hybrid ensemble techniques, specifically leveraging ARIMA error residual modeling alongside a Transformer-based meta-learner to blend base model outputs dynamically.

Extending beyond price target regression, Task C.7 transitions the models into directional classification tasks across five distinct stock tickers while incorporating real-time market sentiment:

1. **Feature Engineering & Baseline Training:** Technical indicators are engineered from OHLCV data. Recurrent architectures (LSTM, GRU, RNN) are trained as baseline directional classifiers using purely technical data.
2. **GNews Extraction & Sentiment Analysis:** Relevant financial headlines and news articles are extracted via the GNews API. Natural Language Processing (NLP) techniques are applied to quantify news articles into normalized numerical sentiment scores.
3. **Multi-Modal Data Integration:** The sentiment scores are merged with the engineered technical feature set to construct a multi-modal dataset.
4. **Comparative Evaluation:** The classifiers are re-trained and benchmarked (comparing baseline performance without sentiment against with sentiment) to isolate and evaluate the impact of news sentiment on financial movement predictions.

2. Overall System Architecture

2.1 Data Flow



2.2 Environment

To successfully replicate and run the stock price prediction model, please fulfill the following structural and software dependencies.

The system is built on an isolated virtual environment (venv) to ensure dependency abstraction and eliminate version conflicts with global system libraries:

- Core Runtime: **Python 3.11.x (64-bit)**
- Execution Interface: **Windows Command Prompt (CMD)**

To reconstruct this environment layout from scratch, run the following sequential commands within the target file directory: **(make sure you have installed Python 3.11)**

```
py -3.11 -m venv venv

venv\Scripts\activate.bat

pip install -r requirements.txt
```

Every time running a python file in this project's folder, please active the virtual environment first using **venv\Scripts\activate.bat**

3. Implemented Data Processing Techniques

Data processing forms the foundation of the predictive framework. Because the system handles two distinct problem formulations, continuous target regression across single/multi-step horizons (From Task C.1 to C.6) and binary directional classification across multiple assets (Task C.7), the data processing pipeline is structured into two specialized modules.

From Task C.1 to C.6, we focused on preparing historical Open-High-Low-Close-Volume (OHLCV) market data for time-series regression and ensembling. All preprocessing operations are encapsulated into a modular pipeline (load_scale_and_split_data) to ensure complete experimental reproducibility and prevent data leakage:

- Caching and Ingestion: Checks local storage (os.path.exists()) before querying yfinance to avoid API rate limits and reduce latency.
- Structure Flattening: Automatically converts multi-index column headers down to 1D arrays to prevent data format errors in machine learning pipelines.
- Missing Data Handling: Uses forward-fill (ffill) to handle missing trading days, preserving unbroken sequence length without introducing future data bias.
- Chronological Splitting: Uses date-based or ratio-based sequential splits (80/20) rather than random splits to prevent look-ahead data leakage.

- Scaler Management: Scalers (MinMaxScaler) are fitted only on training data, transformed across test data, and stored in a dictionary to allow exact dollar-value inverse transformations later.

Task C.7 extends the system from continuous price regression to directional binary classification across five primary Australian Securities Exchange (ASX) equities: CBA, BHP, NAB, CSL, and WBC:

- 18 Technical Indicators: Transforms raw OHLCV market data for 5 ASX tickers into scale-invariant indicators covering momentum (RSI, Return windows), trend (MACD, Moving Average distances), volatility (ATR, Bollinger Bands), and volume (OBV, Volume ratios).
- FinBERT Sentiment Extraction: Collects financial news via the GNews API, passes combined headline text through ProsusAI/finbert, and calculates a continuous net sentiment score.
- Calendar Alignment and Signal Decay: Reindexes continuous news dates to trading days using an exponential decay factor for non-trading periods, smoothed with a 7-day rolling moving average.
- 3D Tensor Formatting: Fits StandardScaler on technical and sentiment features, then formats data into 60-day lookback sliding windows targeting next-day binary market direction.

4. Experimented Machine Learning Techniques

Continuous Regression & Hybrid Ensembling (Tasks C.4–C.6):

- Base Recurrent Architectures: Standard Deep Learning architectures (LSTM, GRU, Vanilla RNN) were trained on 60-day sliding windows of scaled OHLCV features to generate baseline 1-day and k-day continuous price forecasts.
- ARIMA Error Residual Modeling: An ARIMA time-series model was fitted specifically on the prediction residual errors generated by the base neural networks to capture remaining linear trends and autocorrelations in the error terms.
- Transformer Meta-Learner Aggregation: A Transformer-based meta-learner architecture was used as an ensemble layer, dynamically weighting and blending predictions from the base deep learning models and ARIMA residual corrections.

Directional Classification & Multi-Modal Configurations (Task C.7):

- Binary Classification Framework: Recurrent models were repurposed with Sigmoid activation functions to output next-day directional market movement probabilities across 5 ASX tickers (CBA, BHP, NAB, CSL, WBC).
- Architecture Depth Comparisons: Models were evaluated under two structural hyperparameter regimes:
    - config_1_standard: Lightweight single-layer setups
    - config_2_deep: Deeper, regularized setups

- A/B Input Feature Experiments:
  - Baseline Models: Trained exclusively on the 18 engineered technical indicators.
  - Sentiment-Enriched Models: Trained on technical indicators plus FinBERT NLP sentiment features (net score, 7-day decay score, and news presence flags).
- Statistical Hypothesis Testing: Paired t-tests and Wilcoxon signed-rank tests were conducted on out-of-sample test results to statistically evaluate whether incorporating news sentiment provided meaningful performance gains over technical-only baselines.

5. Critical Analysis of the Implementation

5.1 Phase 1: From Task C.1 to C.6

In Phase 1, combining diverse models through a multi-stage hybrid architecture delivered substantial gains over any standalone base model.

- **Complementary Pattern Recognition (DL + ARIMA):** Standalone neural networks (LSTM, GRU, RNN) capture complex non-linear temporal dynamics well, but often struggle with linear trend extrapolation and residual stationarity. Fitting an ARIMA model specifically on the base DL models' prediction residual errors succeeded in capturing linear autocorrelations that the neural networks missed.
- **Dynamic Transformer Meta-Learner:** Rather than relying on simple averaging or static linear regression to combine outputs, using a Transformer meta-learner allowed the system to dynamically weight base model estimates and ARIMA error corrections based on recent price volatility and temporal context.

5.2 Phase 2: Task C.7

In Phase 2, adding FinBERT news sentiment features to technical indicators failed to improve results and, in several configurations, actively degraded classification performance.

- **Majority-Class Mode Collapse:** When news features were introduced, three sentiment-enriched models (standard GRU, standard RNN, and deep GRU) collapsed into predicting positive price movement 100% of the time. This resulted in 0 True Negatives and 100% Recall, effectively ruining their utility as balanced classifiers.
- **High-Noise Signal and API Sampling Bias:** News fetched via retail APIs (GNews) skews heavily toward general public relations and positive corporate coverage. Introducing a high-variance, net-positive sentiment variable into high-dimensional technical feature space created noise that overwhelmed the loss function during gradient descent.
- **Temporal Lag and Market Efficiency:** By the time a news headline is published, scraped, and processed through FinBERT, institutional algorithms have already priced the information into the stock. The sentiment signal was lagged relative to price action, confusing the model when predicting next-day movements.

- **Synthetic Decay Artifacts:** Applying continuous exponential decay across non-trading days generated artificial sentiment values on trading days that had no actual news events, feeding the models misleading continuous signals.
- **Statistical Confirmation:** Formal hypothesis testing confirmed no statistically significant performance gain from incorporating news sentiment.

6. Conclusion

This project successfully developed and evaluated an end-to-end stock market predictive framework, comparing continuous price regression across multi-stage hybrid architectures (Phase 1, Tasks C.1 - C.6) with directional binary classification using integrated sentiment and technical indicators (Phase 2, Task C.7). The empirical results demonstrate a clear divide in effectiveness between structured quantitative ensembling and unweighted external text integration.

For continuous price forecasting, the multi-stage hybrid ensemble proved remarkably effective. By pairing base recurrent neural networks (LSTM, GRU, RNN) with an ARIMA error residual model and aggregating predictions through a Transformer meta-learner, the system successfully captured both complex non-linear trends and linear autocorrelations. This multi-model approach consistently reduced forecast variance, lowering MAE and RMSE across single-day and multi-step prediction horizons compared to standalone base networks.

In contrast, incorporating news sentiment into directional binary classification produced adverse effects. Integrating FinBERT-derived headline sentiment alongside 18 technical indicators introduced significant noise, API sample bias, and temporal lag, ultimately causing several deep learning configurations to suffer from majority-class mode collapse. Formal statistical hypothesis testing confirmed that headline sentiment provided no statistically significant advantage over technical-only baselines.

Ultimately, the project reveals that while multi-model ensembling is highly robust for refining quantitative price dynamics, coarse retail news sentiment introduces high-variance noise that degrades classification performance. Future iterations of this work should focus on replacing daily headline APIs with real-time, intraday news feeds and implementing regularized loss functions (such as Focal Loss) to prevent model collapse when handling multimodal inputs.